The Rule — ***Split the Terms by the Parities That All Gates of the Current Window Keep.*** *Sectors Then Neither Exchange Nor Merge Terms During the Window.*

	Requirement	Measured Evidence	Answer of the Design
R1	Equal Terms Merge Before Truncation	Otherwise 0.1675–0.189 Instead of 0.1531	The Merge Partner Is in the Same Sector
R2	No Coordination While Gates Run	Cost per Gate Grows From 5.9 to 29.8 ms	No Terms Move During a Window
R3	No Pile of Unmerged Terms	Deferred Merge: 2 of 7 Circuits Do Not Fit	Merge at Every Gate, Inside the Sector
R4	The Split Is Decided at Run Time	Fixed Bits: Largest / Average 1.4–4.9	Rows Chosen From a Sample of Live Terms
R5	Capacity Not Tied to GPU Memory	About 100M Terms per GPU	Sectors Are Kept in Host Memory
R6	Small Problems Do Not Lose	0.42 s on 1 GPU, 32–47 s on 4 GPUs	One Sector on One GPU While Terms Are Few